

Đại học quốc gia TP.HCM
Trường đại học công nghệ thông tin



Ngành: Khoa học máy tính

Môn học: IT005.P21.CTTN

Thực hành Lab 6

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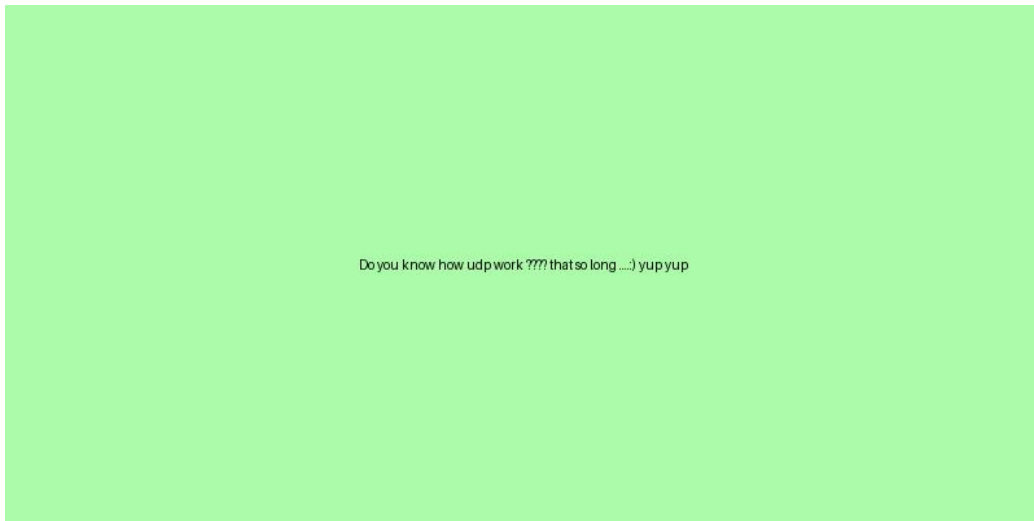


Figure 1: The reconstructed image

1 Task 1 — UDP Image Transmission Packet Capture

1.1 What is the content of the reconstructed image?

The content of the reconstructed image is: "Do you know how udp work ??? that so long:)yupyup"

1.2 How many UDP packets were used in the transmission?

There are 3 UDP packets used in the transmission.

1.3 What is the size of the final image file?

The size of the final image file is 800x400.

2 Task 2 — Subnetting with VLSM and Inter-User Communication

From the perspective of User4, verify full network connectivity by performing ‘ping’ (ICMP echo requests) to all other users in the subnetted network.

2.1 What is the result of each ping attempt?

Table 1: Ping Test Statistics

IP Address	Packets Sent	Packets Received	Packet Loss (%)
172.31.103.30	4	0	100
172.31.103.62	4	0	100
172.31.103.78	4	0	100

2.2 Are there any unreachable hosts? If so, explain the reason.

No one. All three machines cannot be pinged from PC4 because PC4 has not been configured.

2.3 Question 3: Provide the routing path or decision for each successful ping from User4 to the others.

Path from User4 to User3 (172.31.103.78)

This communication is local to the Remote-Site2 router.

1. **User4** sends the packet to its default gateway, Remote-Site2 (at IP 172.31.103.81).

2. **Remote-Site2** checks its routing table. It sees that the destination IP 172.31.103.78 belongs to the 172.31.103.64/28 subnet, which is a **directly connected network** on its GigabitEthernet0/0 interface.
3. **Remote-Site2** forwards the packet directly out of its GigabitEthernet0/0 interface to **User3**. The WAN link is not used for this communication.

Path from User4 to User1 (172.31.103.30)

This communication must cross the WAN link.

1. **User4** sends the packet to its gateway, **Remote-Site2**.
2. **Remote-Site2** checks its routing table and finds a match with the static route: `ip route 172.31.103.0 255.255.255.224 172.31.103.97`.
3. The packet is forwarded over the serial WAN link to the next-hop address 172.31.103.97, which is the **Serial0/0/0** interface of **Remote-Site1**.
4. **Remote-Site1** receives the packet. It sees that the destination IP 172.31.103.30 belongs to a **directly connected network** on its GigabitEthernet0/0 interface.
5. **Remote-Site1** forwards the packet directly to **User1**.

Path from User4 to User2 (172.31.103.62)

This communication also crosses the WAN link.

1. **User4** sends the packet to its gateway, **Remote-Site2**.
2. **Remote-Site2** checks its routing table and finds a match with the static route: `ip route 172.31.103.32 255.255.255.224 172.31.103.97`.
3. The packet is forwarded over the serial WAN link to the next-hop address 172.31.103.97 (**Remote-Site1**).
4. **Remote-Site1** receives the packet. It sees that the destination IP 172.31.103.62 belongs to a **directly connected network** on its GigabitEthernet0/1 interface.
5. **Remote-Site1** forwards the packet directly to **User2**.

2.4 Is there any routing misconfiguration that affects communication?

No, there is no routing misconfiguration. Both routers have the correct static routes defined to reach the subnets located on the opposite side of the WAN link.

- **Remote-Site1** correctly knows how to reach the networks of User3 and User4.
- **Remote-Site2** correctly knows how to reach the networks of User1 and User2.